Fiscal Year:	FY 2022	**Task Last Updated:**	FY 12/23/2022
PI Name:	Shepherd, John Ph.D.		
Project Title:	Space-Feasible Body Composition and Body Shape Analysis for Long-Duration Missions II		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	96813	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	TRISH--EXPAND
Start Date:	05/01/2022	**End Date:**	04/30/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wong, Michael Ph.D. (University of Hawaii) Bennett, Johnathan M.S. (University of Hawaii)		
Grant/Contract No.:	NNX16AO69A-EXP0012		
Performance Goal No.:			
Performance Goal Text:			
	This is an EXPAND Project. The long-term goal of our Translational Research Institute for Space Health (TRISH) EXPAND project is to provide accurate multi-compartment body composition assessments to astronauts on long-duration spaceflights to evaluate muscle loss and fat accumulation associated with inactivity and exposure to microgravity. Our overall approach is to recalibrate 4-compartment models of body composition derived in standard gravity on large populations using accessible 3D optical (3DO) body scanning and bioimpedance analysis (BIA) measures, for space conditions where hydration and body fluid distributions are altered. Our central hypothesis is that alterations to whole-body shape due to microgravity-induced fluid redistribution is predictable from pre-flight body shape and composition, such that the accuracy of in-flight body composition models will remain unchanged. Our long-term vision is to integrate our in-flight		

Task Description:	body composition models with dietary, pharmacological, and physical activity interventions to reduce catastrophic health events such as bone fractures, acute fatigue, or metabolic instability (diabetes, heart disease, etc.). During this funding period, we plan the following specific aims to be completed as part of the SpaceX Polaris Mission: 1. Identify the specific combination of 3DO and BIA components needed to satisfy flight readiness, accuracy, and precision of body composition measures. 2. Describe the characteristics of body shape changes due to microgravity using pre-flight and day 1 scans. 3. Describe the test-retest precision and least significant change in 3DO total and regional body composition and anthropometry measurements pre/post flight and utilize this knowledge to improve the accuracy of body composition modeling. 4. Contrast the change in criterion body composition measures immediately before and after flight to similar measures acquired in-flight for the first and last day of the mission.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2022.
Bibliography Type:	Description: (Last Updated: 12/27/2024)